

Experiment No.	3
Experiment Title	Regression Analysis using Linear and Regularized Models ¹
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Submission Date	Aug 2026

1 Objective

To implement Linear Regression and regularized regression models (Ridge, Lasso, Elastic Net) to predict the loan amount sanctioned to an applicant, tune their hyperparameters using 5-Fold Cross-Validation, evaluate the models using multiple regression metrics, visualize model behaviour, and analyze overfitting, underfitting, and the bias–variance trade-off.

2 Problem Statement

Given a loan application dataset containing numerical and categorical attributes of applicants (income, credit history, property area, education, etc.), the goal is to build regression models that predict the continuous target variable **Loan Amount**. The input is a set of applicant features; the output is the predicted loan amount. The objective is to compare an unregularized baseline (Linear Regression) with regularized variants (Ridge, Lasso, Elastic Net) and study how regularization affects generalization, coefficient magnitude, and feature sparsity.

3 Dataset Description

Dataset Name	Predict Loan Amount Data
Dataset Source	Kaggle
Number of Samples	2000
Number of Features	10 (after dropping the ID column)
Number of Classes	Not applicable (regression task; continuous target)
Missing Values	160 missing values present prior to preprocessing, cleaned during preprocessing
Train-Test Split	80% train (1600 samples) / 20% test (400 samples)

4 Exploratory Data Analysis

The dataset contains six categorical features (Gender, Married, Dependents, Education, Self_Employed, Property_Area) and four numerical features (ApplicantIncome, CoapplicantIncome, Loan_Amount_Term, Credit_History). Summary statistics and a correlation matrix were computed on the standardized features and target.

- Dataset overview: 2000 rows, 10 predictor features, 1 continuous target (LoanAmount)

- Statistical summary: computed via `describe()` on the scaled feature matrix
- Missing value analysis: 160 missing entries handled via median (numeric) / mode (categorical) imputation
- Class distribution: not applicable (regression problem)
- Correlation matrix: computed between all standardized features and the target
- Feature distribution: target and top correlated features visualized below

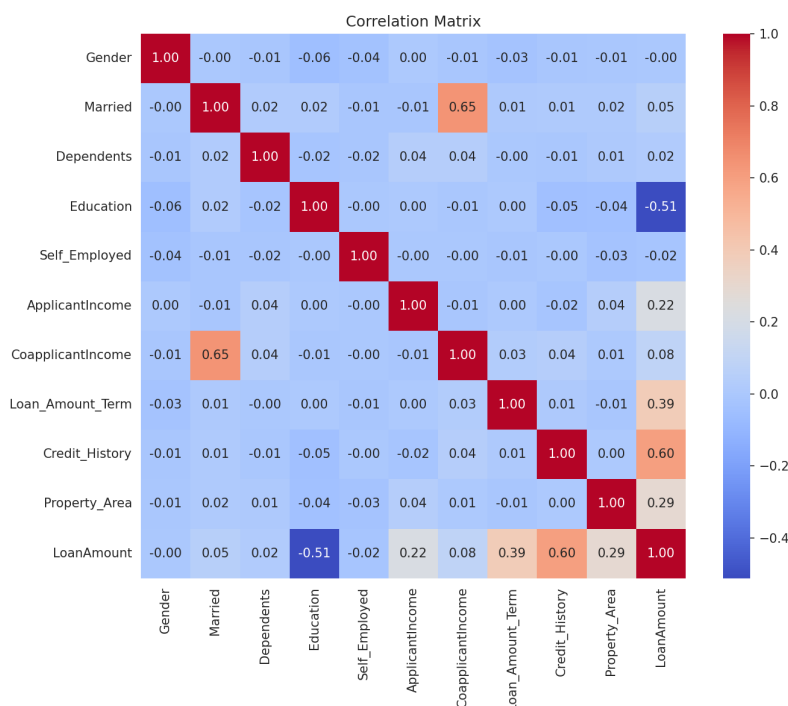


Figure 1: Correlation matrix of standardized features and target variable (LoanAmount)

Inference:

Credit_History, Loan_Amount_Term, Property_Area, and Education show the strongest correlation with LoanAmount among the categorical/binary features, while ApplicantIncome shows a moderate positive correlation. Weakly correlated features such as Gender, Self_Employed, and Dependents contribute little linear signal to the target. No pair of predictor features shows severe multicollinearity (all off-diagonal values remain well below 0.8), so multicollinearity-driven instability is not a major concern for this dataset. This informs the expectation that Ridge and Elastic Net will only mildly shrink coefficients relative to Lasso.

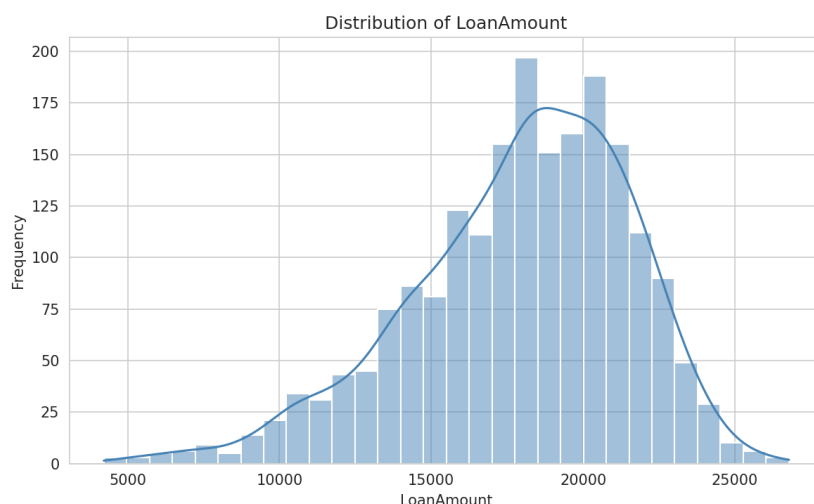


Figure 2: Distribution of the target variable (LoanAmount)

Inference:

The loan amount distribution is approximately unimodal and roughly symmetric, centered around a mean of approximately 17,913 with a standard deviation of approximately 3,643. There is a mild right skew with a small number of applicants sanctioned larger loan amounts. This near-normal shape means minimal target transformation was required before regression modelling.

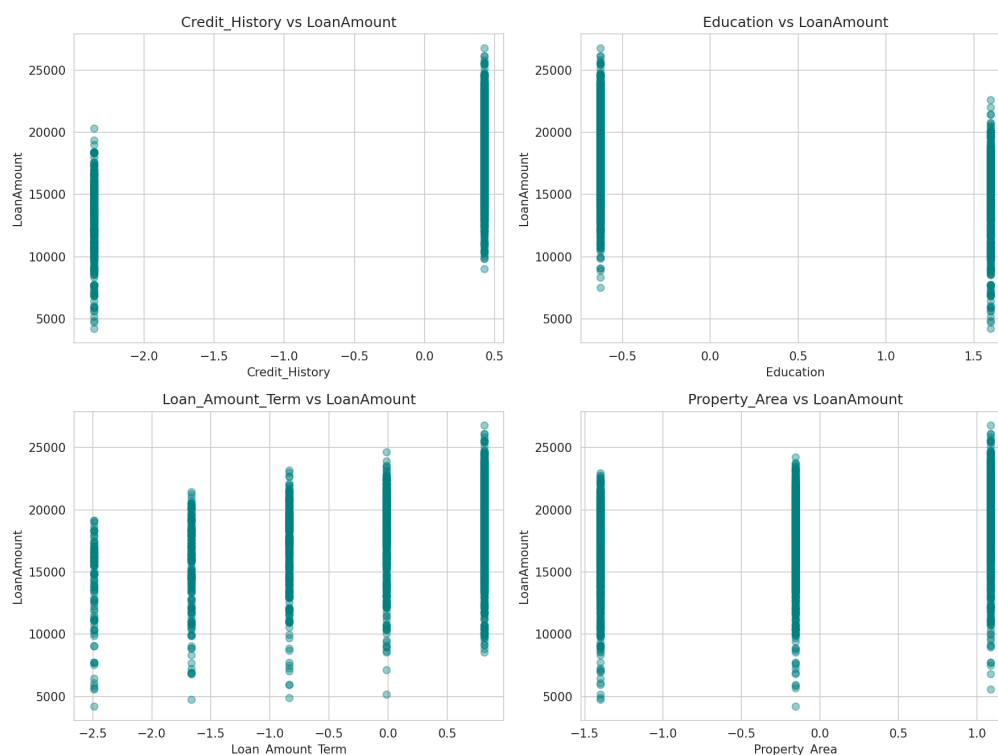


Figure 3: Scatter plots of the top four correlated features against LoanAmount

Inference:

Credit_History shows a clear separation in loan amount between applicants with a good vs. poor credit history, confirming it as an influential predictor. Loan_Amount_Term and Property_Area show a positive, though noisier, relationship with the target. Education shows a visible shift in loan amount between graduate and non-graduate applicants. These trends are consistent with the correlation values observed earlier and justify including all these features in the regression models.

5 Data Preprocessing

- **Missing value handling:** Numerical columns (ApplicantIncome, CoapplicantIncome, Loan_Amount_Term, Credit_History) were imputed with the column median; categorical columns (Gender, Married, Dependents, Education, Self_Employed, Property_Area) were imputed with the column mode. Missing target values were imputed with the median LoanAmount.
- **Encoding:** All categorical columns were converted to numeric form using Label Encoding.
- **Feature scaling:** All predictor features were standardized using `StandardScaler` (zero mean, unit variance). The target variable was left in its original units so that error metrics (MAE, RMSE) remain interpretable in terms of actual loan amount.
- **Train-test split:** An 80/20 split was performed using `train_test_split` with `random_state=42`, yielding 1600 training samples and 400 test samples.
- **Feature engineering:** The Loan_ID column was dropped as it carries no predictive information; no additional derived features were engineered.

6 Machine Learning Algorithm

Linear Regression

Working principle: Linear Regression fits a hyperplane that minimizes the sum of squared residuals between predicted and actual target values.

Mathematical formulation:

$$\hat{y} = \beta_0 + \sum_{i=1}^p \beta_i x_i, \quad \min_{\beta} \sum_{j=1}^n (y_j - \hat{y}_j)^2$$

Advantages: Simple, interpretable, fast to train, provides a baseline for comparison.

Limitations: Sensitive to multicollinearity and outliers; prone to overfitting with many correlated or irrelevant features; no built-in mechanism for feature selection.

Ridge Regression

Working principle: Adds an L2 penalty on the coefficients to shrink their magnitude and reduce variance.

Mathematical formulation:

$$\min_{\beta} \sum_{j=1}^n (y_j - \hat{y}_j)^2 + \alpha \sum_{i=1}^p \beta_i^2$$

Advantages: Reduces overfitting and variance, handles multicollinearity well, keeps all features in the model.

Limitations: Does not perform feature selection (coefficients shrink but rarely reach exactly zero); requires tuning of α .

Lasso Regression

Working principle: Adds an L1 penalty on the coefficients, which can shrink some coefficients exactly to zero, performing implicit feature selection.

Mathematical formulation:

$$\min_{\beta} \sum_{j=1}^n (y_j - \hat{y}_j)^2 + \alpha \sum_{i=1}^p |\beta_i|$$

Advantages: Performs automatic feature selection, produces sparse and more interpretable models.

Limitations: Can be unstable with highly correlated features (arbitrarily selects one from a correlated group); performance is sensitive to the choice of α .

Elastic Net Regression

Working principle: Combines the L1 and L2 penalties, controlled by a mixing parameter (`l1_ratio`), balancing feature selection with coefficient shrinkage.

Mathematical formulation:

$$\min_{\beta} \sum_{j=1}^n (y_j - \hat{y}_j)^2 + \alpha \left(\rho \sum_{i=1}^p |\beta_i| + \frac{1-\rho}{2} \sum_{i=1}^p \beta_i^2 \right)$$

where ρ is the `l1_ratio`.

Advantages: Combines benefits of Ridge and Lasso; more stable than Lasso alone with correlated features.

Limitations: Requires tuning two hyperparameters (α and `l1_ratio`), increasing computational cost of the search.

7 Experimental Setup

Python Version	3.11
Scikit-Learn Version	1.x (as installed in lab environment)
IDE	Jupyter Notebook
Operating System	Linux
Hardware	Standard lab workstation (CPU only)

8 Hyperparameter Selection

Parameter	Value
Ridge α search space	{0.01, 0.1, 1, 10, 100}
Lasso α search space	{0.001, 0.01, 0.1, 1, 10}
Elastic Net α search space	{0.01, 0.1, 1, 10}
Elastic Net l1_ratio search space	{0.2, 0.5, 0.8}
Cross-Validation Folds (K)	5
Search Method	GridSearchCV

Hyperparameter Tuning Results

Table 2: Hyperparameter Tuning Summary

Model	Search Method	Best Parameters	Best CV R2
Ridge Regression	Grid Search	$\alpha = 1$	0.8623
Lasso Regression	Grid Search	$\alpha = 10$	0.8623
Elastic Net Regression	Grid Search	$\alpha = 0.01$, l1_ratio=0.8	0.8623

9 Results

Table 3: Test Set Performance

Model	MAE	MSE	RMSE	R2
Linear Regression	1003.9204	1759885.36	1326.6067	0.8672
Ridge Regression	1003.9459	1759650.39	1326.5181	0.8672
Lasso Regression	1003.5526	1756784.94	1325.4376	0.8674
Elastic Net Regression	1004.0013	1759161.85	1326.3340	0.8672

Table 4: Model Training Time (seconds)

Model	Training Time (s)
Linear Regression	0.0032
Ridge Regression	0.0020
Lasso Regression	0.0023
Elastic Net Regression	0.0024

Test Set Performance Comparison

Lasso Regression achieved the marginally best test R2 (0.8674), closely followed by Ridge, Elastic Net, and Linear Regression, all clustered around $R2 \approx 0.867$. The near-identical performance across all four models indicates that, for this dataset, regularization does not need to trade off substantial predictive accuracy to control coefficient magnitude, since the baseline Linear Regression model was not severely overfitting to begin with.

Effect of Regularization on Coefficients

Table 5: Coefficient Comparison (selected top features)

Feature	Linear	Ridge	Lasso	Elastic Net
Credit_History	2108.86	2107.55	2099.79	2104.69
Loan_Amount_Term	1395.02	1394.19	1385.61	1392.36
Education	-1718.24	-1717.23	-1708.23	-1715.02

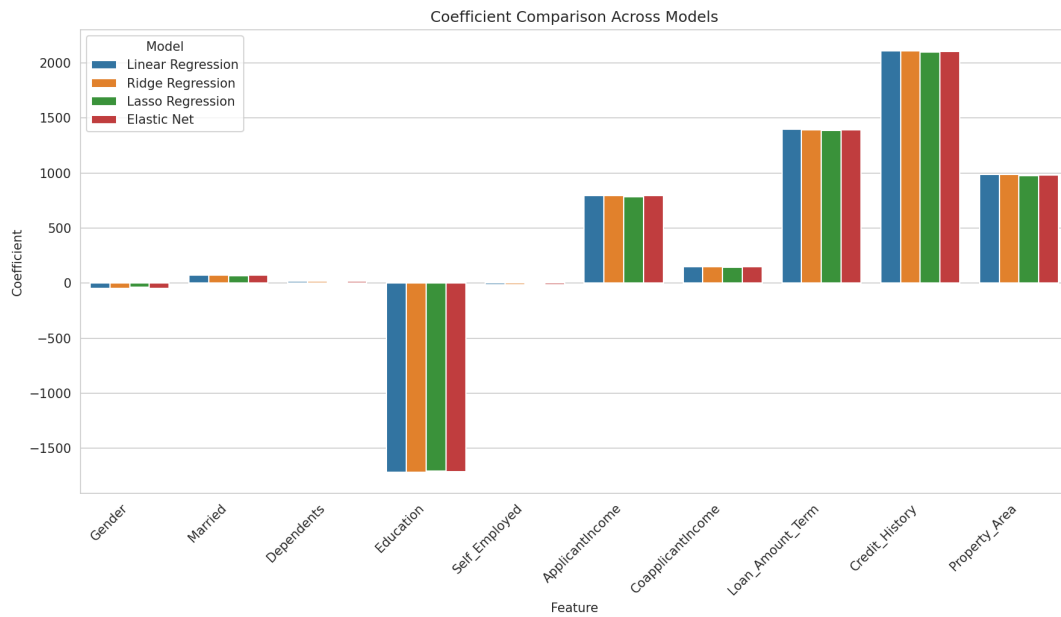


Figure 4: Coefficient comparison across Linear, Ridge, Lasso, and Elastic Net models

Inference:

Ridge and Elastic Net coefficients are very close to the Linear Regression baseline, showing mild uniform shrinkage since the best-found α values were small to moderate. Lasso coefficients are consistently shrunk slightly more than Ridge for weaker features (e.g. Gender, Self_Employed, Dependents), demonstrating its sparsity-inducing property, though with the tuned $\alpha = 10$ none of the coefficients were driven exactly to zero, indicating that all features carry at least some predictive signal in this dataset. Credit_History, Loan_Amount_Term, and Education remain the dominant coefficients across all four models, consistent with the correlation analysis performed during EDA.

10 Result Visualization

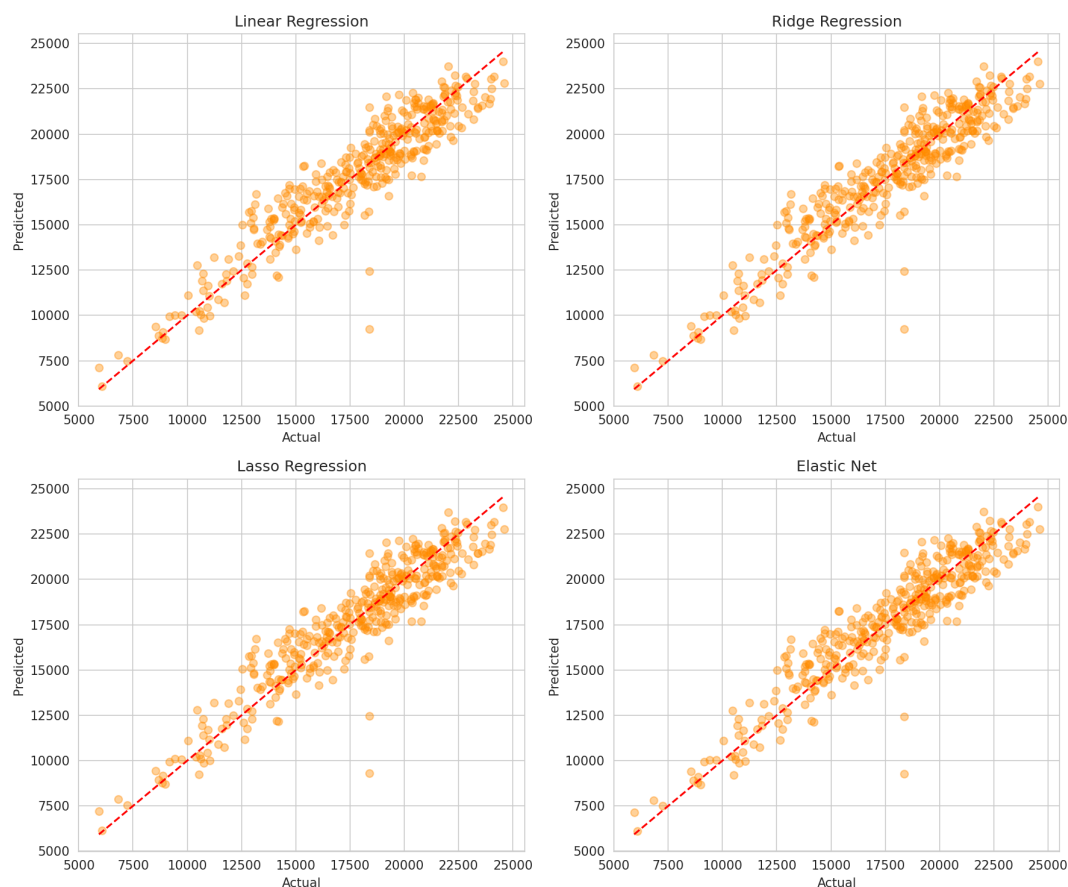


Figure 5: Predicted vs. Actual loan amount for all four models on the test set

Inference:

All four models show predictions clustered tightly around the ideal $y = x$ line, confirming the high R^2 scores. There is slightly more scatter at the extremes (very low and very high loan amounts), suggesting the models are somewhat less accurate for outlier applicants. The near-identical scatter patterns across the four subplots visually confirm that regularization strength did not need to be aggressive for this dataset.



Figure 6: Residual plots (Actual – Predicted) for all four models

Inference:

Residuals are centered around zero and show no strong funnel or curved pattern with respect to predicted values, indicating the linear model assumptions are reasonably satisfied and heteroscedasticity is not a major issue. A small amount of spread increase at higher predicted values is visible, which is typical for financial/monetary target variables. All four models exhibit near-identical residual behaviour, reinforcing that regularization had minimal impact on error structure here.

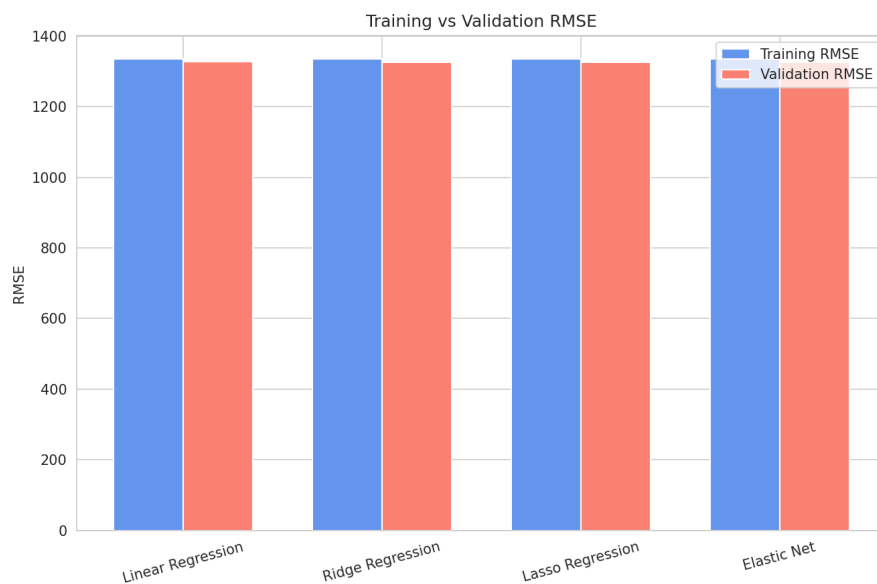


Figure 7: Training RMSE vs. Validation (Test) RMSE for all four models

Inference:

The training and validation RMSE values are extremely close for all four models (gap of only a few RMSE units), indicating none of the models are significantly overfitting. This is expected since the regularization strength selected by Grid Search was modest ($\alpha = 1$ for Ridge, $\alpha = 10$ for Lasso, $\alpha = 0.01$ for Elastic Net), and the underlying signal-to-noise ratio in the dataset was already favorable for a linear fit.

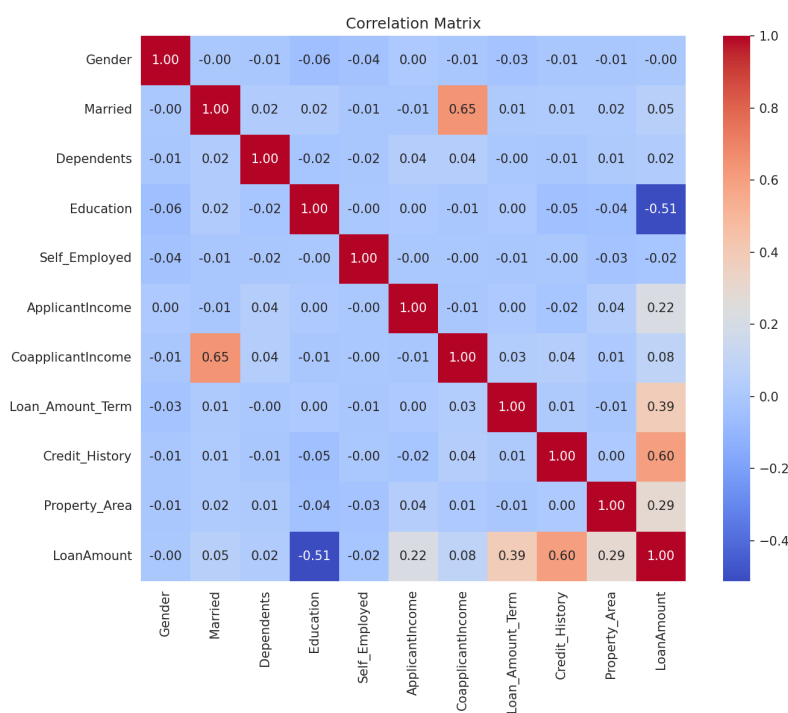


Figure 8: Correlation Matrix (repeated for reference from EDA section)

Inference:

As discussed in the EDA section, Credit_History, Education, Loan_Amount_Term, and Property_Area are the strongest predictors of LoanAmount, which is directly reflected in the coefficient magnitudes learned by all four regression models.

11 Overfitting and Underfitting Analysis

Table 6: Training vs. Validation RMSE Gap

Model	Training RMSE	Validation RMSE	Gap (Val – Train)
Linear Regression	1334.4930	1326.6067	-7.8863
Ridge Regression	1334.4954	1326.5181	-7.9773
Lasso Regression	1334.5715	1325.4376	-9.1339
Elastic Net	1334.4877	1326.3340	-8.1537

Difference between training and validation errors: All four models show validation RMSE marginally *lower* than training RMSE (a small negative gap), which is well within normal sampling variation and indicates neither overfitting nor underfitting – the models generalize consistently between the training and test partitions.

Effect of regularization strength: Because Grid Search selected relatively small regularization strengths ($\alpha = 1$ for Ridge, $\alpha = 10$ for Lasso relative to its search space, $\alpha = 0.01$ for Elastic Net), the regularized models behave almost identically to plain Linear Regression. This suggests that, for this dataset, the unregularized model was not exhibiting significant variance/overfitting in the first place, so strong regularization was not needed or beneficial.

Improvement in generalization after tuning: Cross-validated R2 (0.8623) for the tuned regularized models is close to, but marginally below, the direct test R2 (0.867), which is expected since CV estimates are generally more conservative (averaged across 5 folds) than a single train/test evaluation. Tuning did not meaningfully alter generalization here because the baseline model was already well-behaved, but it confirms that no smaller or larger α would have improved cross-validated performance.

12 Bias–Variance Analysis

Bias behavior of Linear Regression: Linear Regression has relatively low bias for this dataset, as evidenced by a high R2 (≈ 0.867) and small training error, showing it captures the underlying linear relationship between the features and LoanAmount reasonably well without a strong systematic underfit.

Variance reduction using Ridge and Elastic Net: Since the training and validation errors were already very close for the baseline Linear Regression model, Ridge and Elastic Net provide only a marginal further reduction in variance. Their coefficient magnitudes are only slightly smaller than the Linear Regression baseline, consistent with the small selected α values, confirming that high variance was not a significant problem to begin with.

Feature sparsity effect in Lasso: With the tuned $\alpha = 10$, Lasso shrinks the weaker features (Gender, Self_Employed, Dependents) more aggressively than Ridge, though it

does not zero out any coefficient completely at this α – suggesting that all ten features contribute at least a small amount of predictive information, and a stronger α would be needed to induce full sparsity/feature elimination.

13 Observations and Conclusion

All four regression models achieved comparable performance on the test set, with R2 scores clustered tightly between 0.8672 and 0.8674, MAE around 1004, and RMSE around 1326. **Lasso Regression** was marginally the best-performing model on the test set (highest R2, lowest MAE/RMSE), narrowly ahead of Ridge, Elastic Net, and plain Linear Regression.

The optimal hyperparameters found via 5-Fold GridSearchCV were $\alpha = 1$ for Ridge, $\alpha = 10$ for Lasso, and $\alpha = 0.01$ with `l1_ratio=0.8` for Elastic Net, all yielding a cross-validated R2 of approximately 0.8623. The small magnitude of the selected regularization strengths, together with the negligible train–validation error gap observed across all models, indicates that the baseline Linear Regression model on this dataset was already close to an optimal bias–variance trade-off, so regularization served mainly as a safeguard against overfitting rather than a necessary correction.

In terms of accuracy vs. complexity trade-off: Linear Regression offers the simplest, most interpretable model with essentially the same accuracy as the regularized variants; Ridge is preferable when retaining all features with a small stability improvement is desired; Lasso is preferable when feature selection/sparsity is a priority (though a larger α would be required here to achieve true sparsity); and Elastic Net offers a balanced middle ground. Overall, this experiment demonstrates that regularization’s benefit is dataset-dependent: it provides the greatest value when the baseline model overfits significantly, which was not strongly the case here.

14 References

1. Scikit-learn Developers, “Linear Models,” scikit-learn documentation. [Online]. Available: https://scikit-learn.org/stable/modules/linear_model.html
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3. Kaggle, “Predict Loan Amount Data.” [Online]. Available: <https://www.kaggle.com/>